MODULE 12, LESSON 3

AI 101 FOR MECHANICAL ENGINEERS: WINDOWS AND HORIZONS

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Structure and Outline

1. Foundational Timing Quantities
2. Signal Windows and Horizons
3. Timeline Construction and Availability
4. Filtering and Timestamp Constraints

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Section 1
Foundational Timing Quantities

Purpose: To define an exact taxonomy of time-series parameters, distinguishing physical time metrics from data matrix metrics.

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Time-Series Parameter Matrix

<table>
<tr><th>Category 1: Physical Time Metrics</th><th>Category 2: Data Matrix Metrics</th></tr>
<tr><td>Sampling Rate: Frequency of samples.</td><td rowspan="2">Filtered Row Lag: Matrix index shift.
Requires timestamp verification.</td></tr>
<tr><td>Forecast Horizon: Issue-to-target time interval.</td></tr>
<tr><td>Window Length: Period summarized by features.</td><td rowspan="2">Time Series: Observations indexed by time.</td></tr>
<tr><td>Alignment: Time associated with each window.</td></tr>
</table>

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Section 2

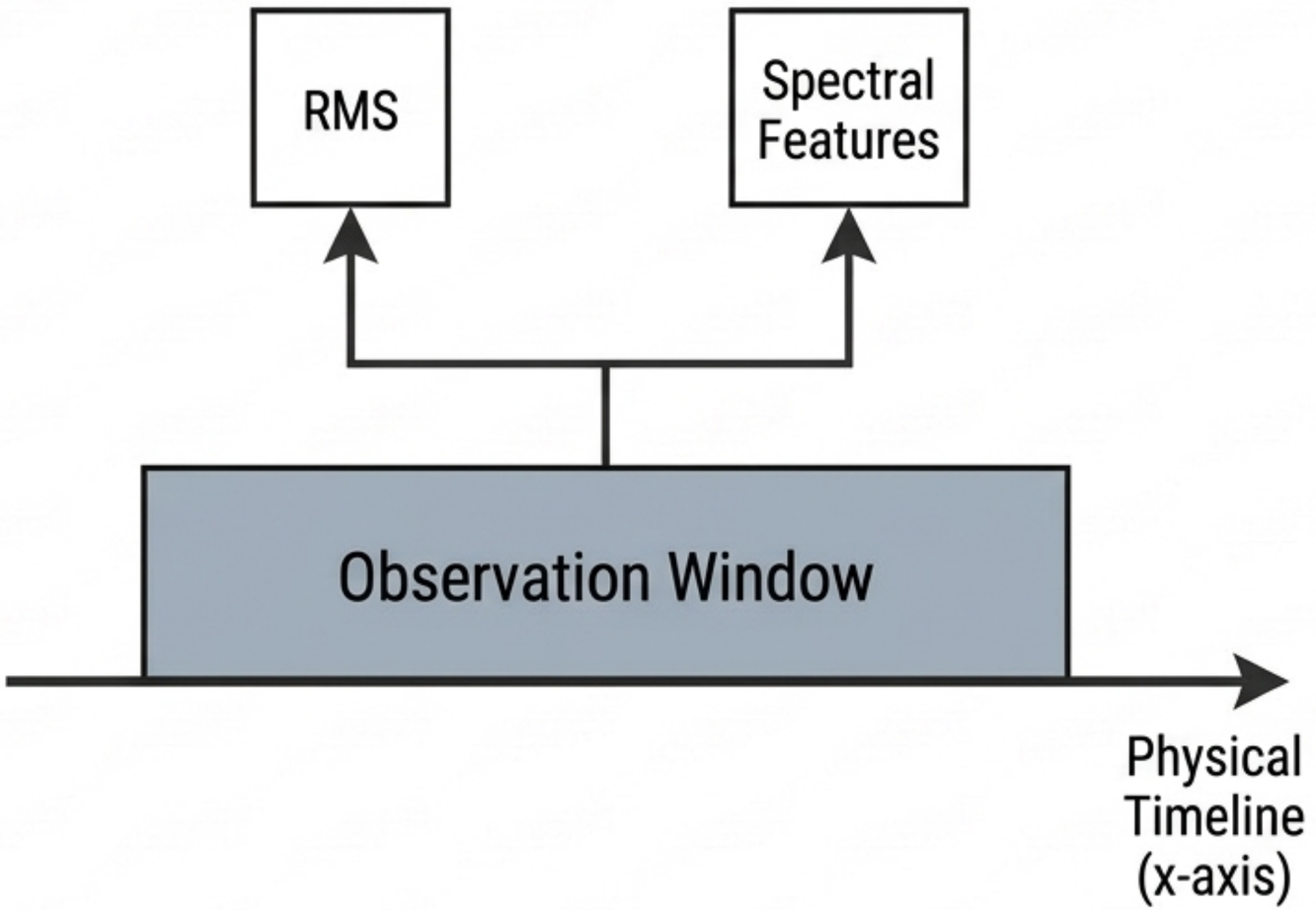
Signal Windows and Horizons

Purpose: To geometrically model the time intervals required for feature extraction and forecasting.

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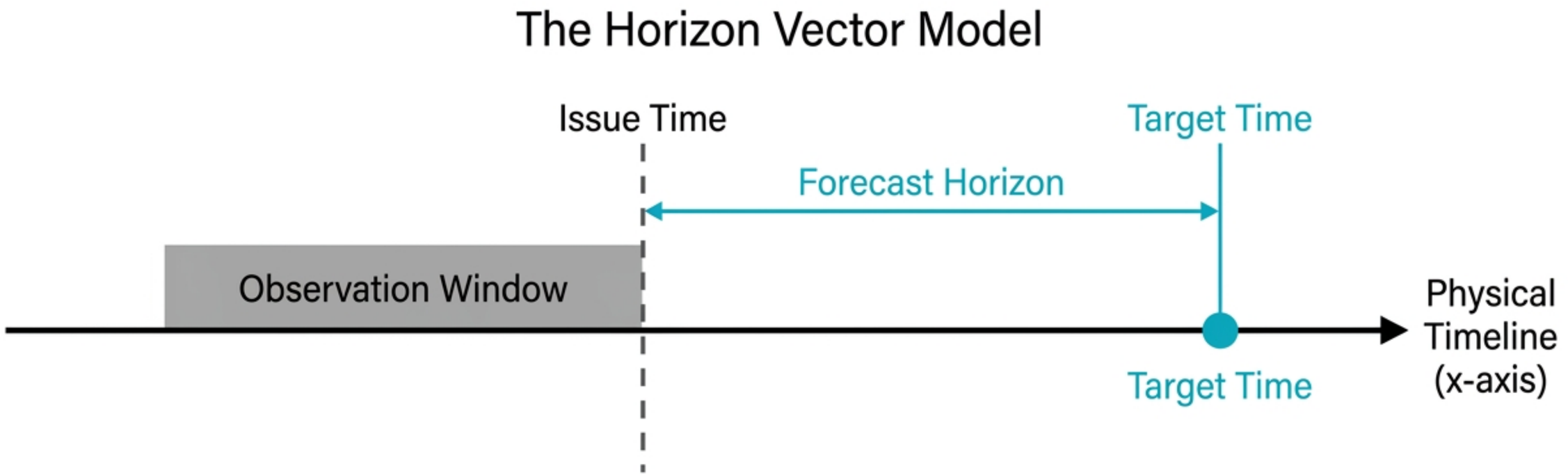
Signal Window Feature Extraction

- Signal features are computed over strictly defined temporal windows.
- RMS summarizes magnitude within the window limit.
- Spectral features summarize frequency content.
- Always state sampling rate, window length, and alignment.



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The Forecast Horizon

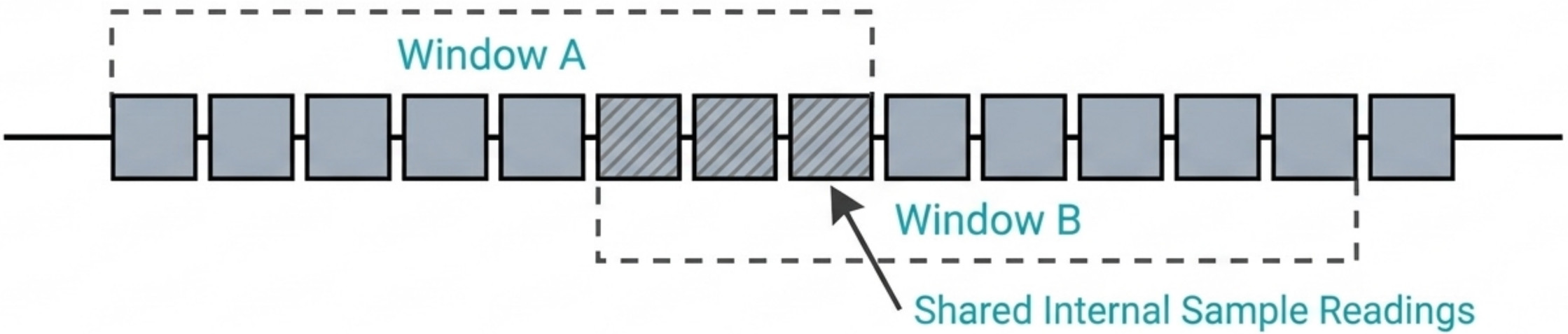


- Horizon is the time between issue and target.
- Real-time detector features cannot include future readings.
- Example: A day-ahead prediction defines a 24-hour horizon.

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Overlapping Window Split Constraints

Overlapping Split Schematic



- Overlapping windows require careful temporal splits.
- Neighbouring windows frequently share internal sample readings.
- Leave suitable gaps when the task requires independence.
- Hold out discrete events to maintain valid test boundaries.

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Section 3

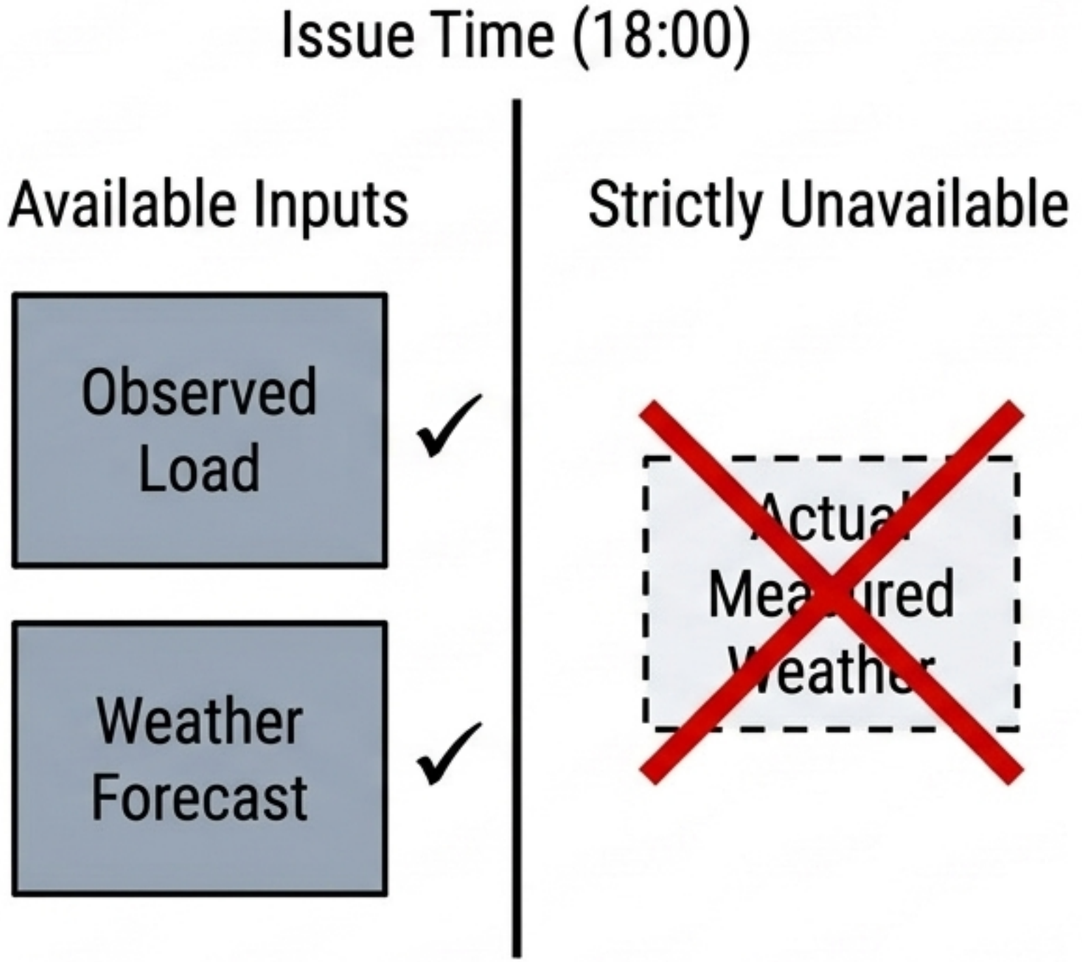
Timeline Construction and Availability

Purpose: *To establish operational boundary conditions for sequence predictors based on chronological availability.*

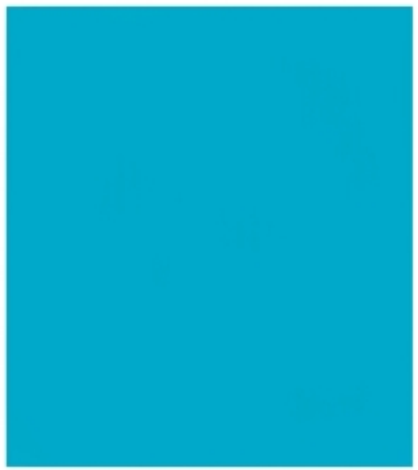
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Chronological Data Availability

- Draw timelines of available observations before fitting predictors.
- System load up to the issue time is available.
- Issued weather forecasts are valid inputs with stated uncertainty.
- Future actual measured weather is strictly unavailable.
- Using future data evaluates a false information scenario.



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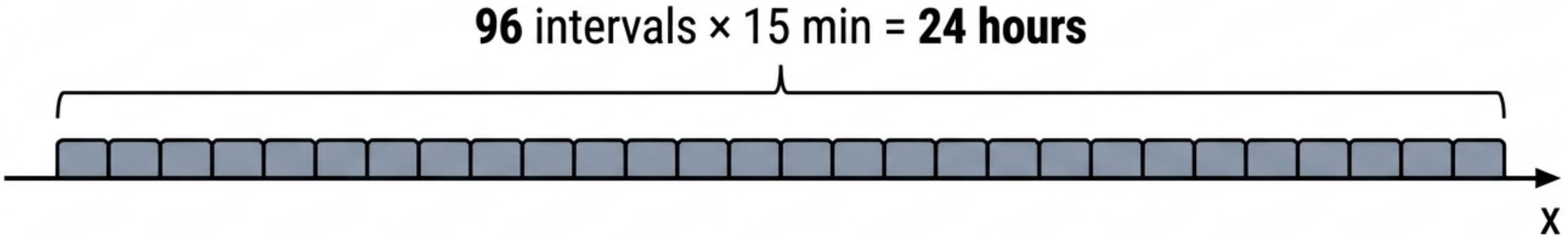


Section 4
Filtering and Timestamp Constraints

Purpose: To systematically demonstrate the failure of row-based shifting and formally establish the Timestamp Contract.

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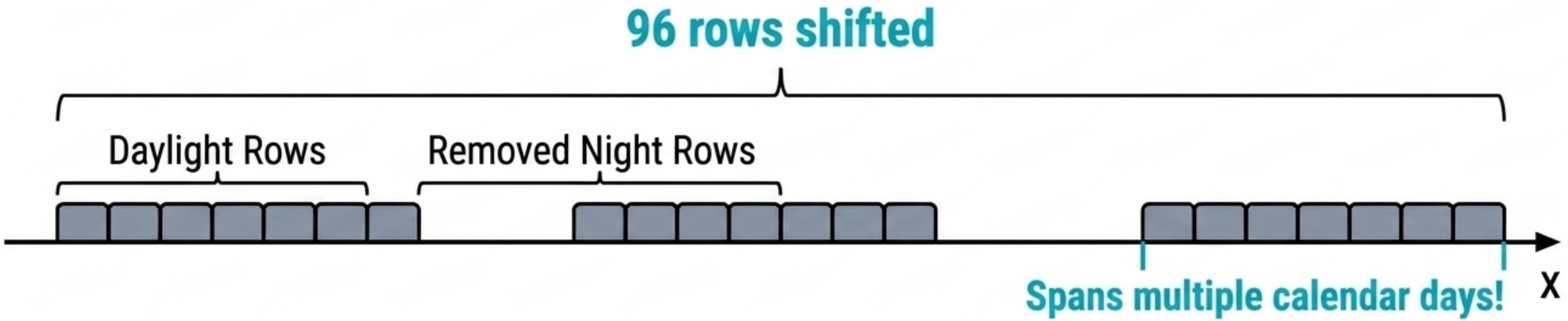
Complete Regular Data Constraints



- Ninety-six 15-minute intervals theoretically equal 24 hours.
- This calculation assumes complete, regular, uninterrupted sample data.
- Formula: 24 [h] × 60 [min/h] / 15 [min/sample] = 96 samples.
- In this ideal state, positional row shifts match physical time.

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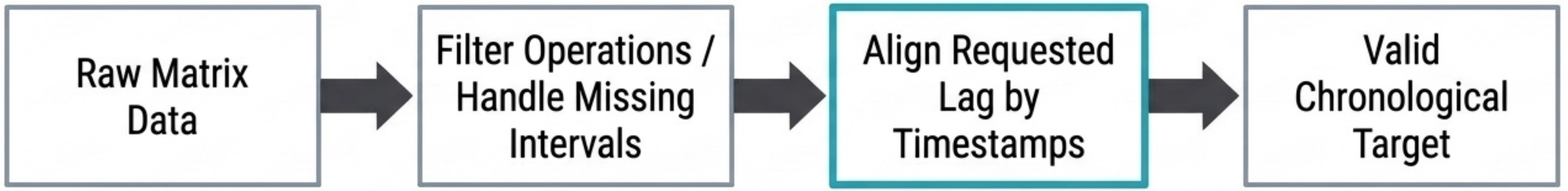
The Filtered Matrix Failure Mode



- Filtering significantly alters positional time spacing.
- Removing night rows destroys the regular chronological interval.
- After filtering, a 96-row shift spans multiple calendar days.
- Filtered row positions cannot substitute for chronological timestamps.

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The Timestamp Contract



- Always align the requested lag and target by chronological timestamps.
- Record and account for all missing intervals explicitly.
- Describe how changed sampling rates are handled programmatically.
- Test models strictly chronologically to prevent data leakage.

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Key Points

- Foundational parameters distinguish physical time metrics from matrix indices.
- Signal windows and forecast horizons require strict geometric boundaries.
- Timelines dictate real-world model constraints based on issue-time availability.
- The timestamp contract supersedes row positions when handling filtered series.

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